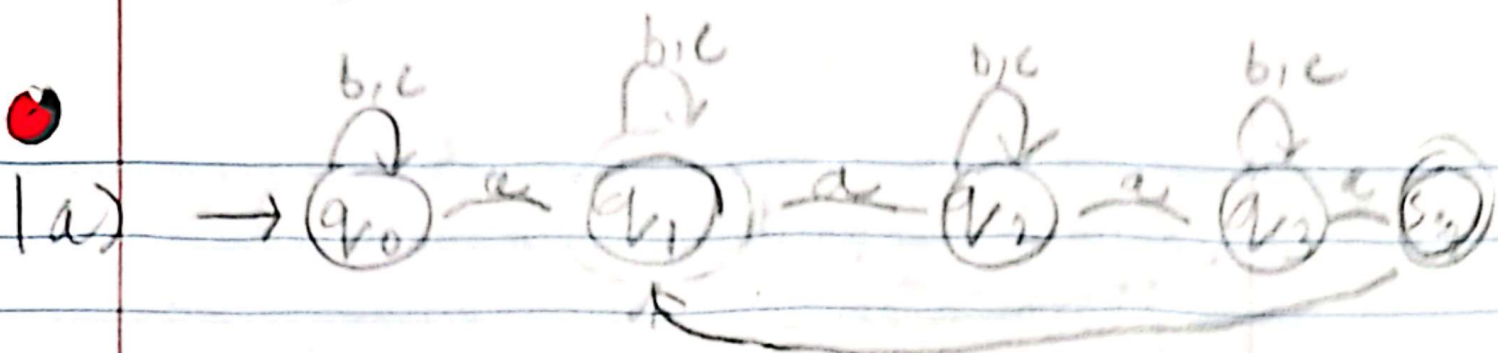




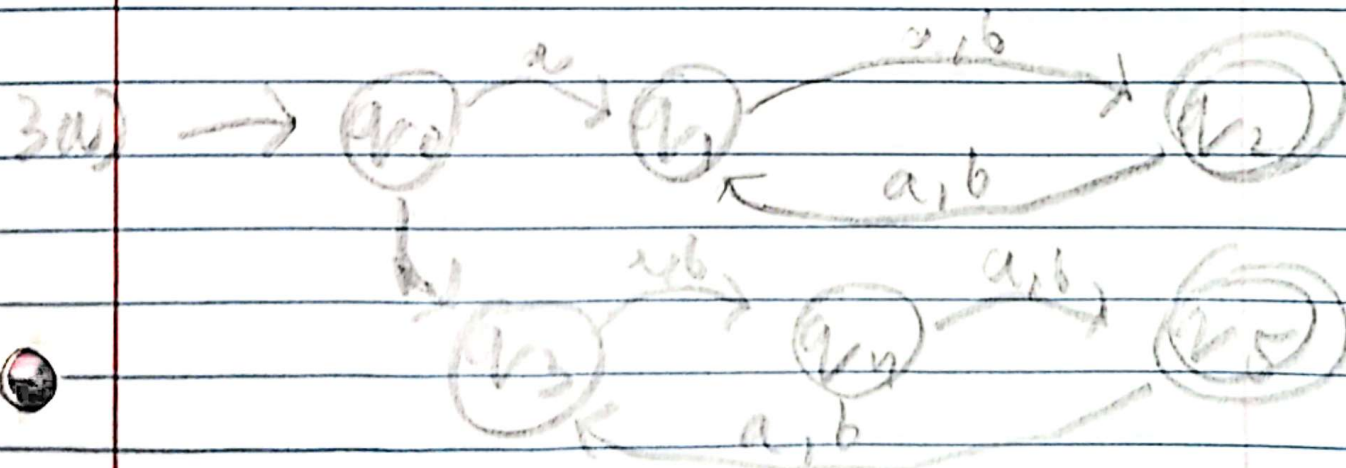
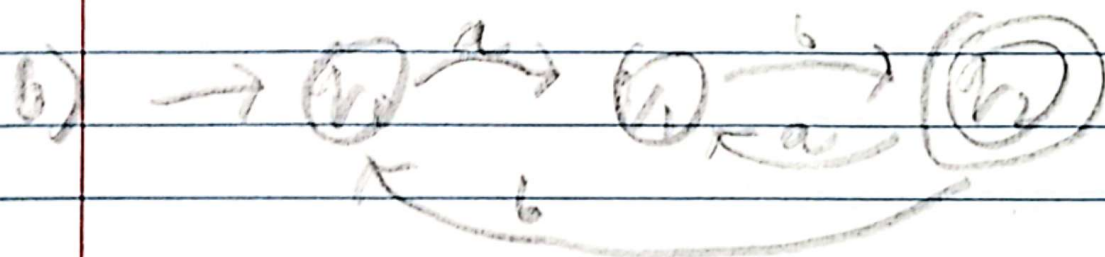
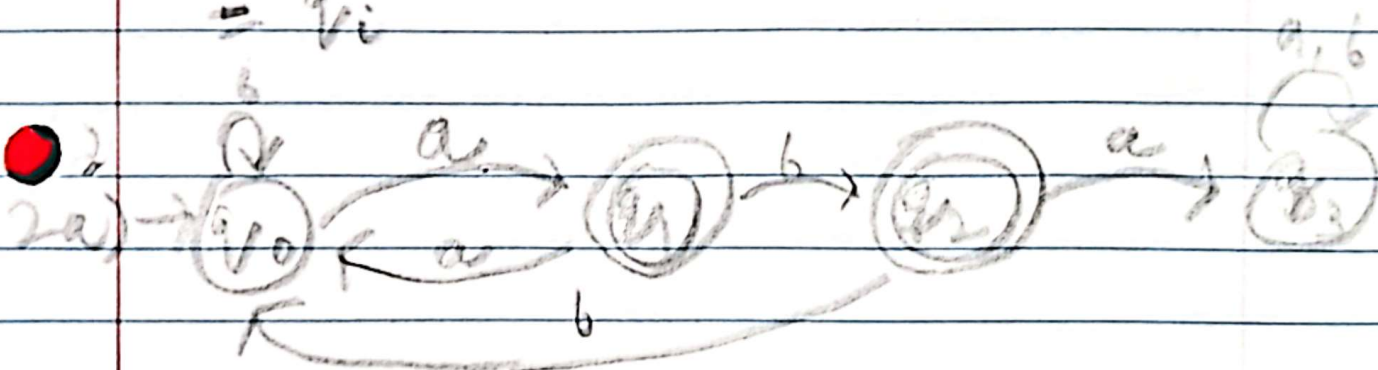
b) Alphabet: $\Sigma = \{a, b, c\}$

States: $Q = \{q_0, q_1, \dots, q_{k-1}\}$

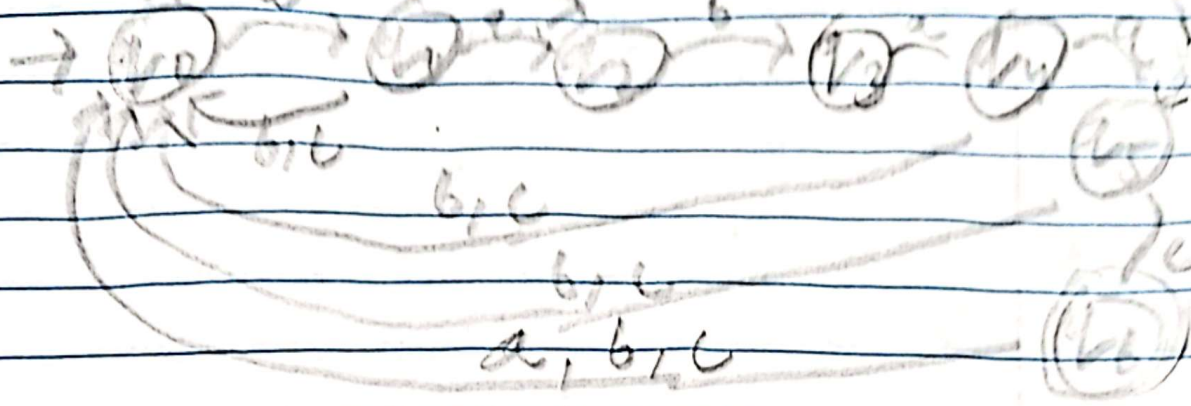
Start State: $s_0 = q_0$

Accept States: $F = \{q_k\}$

Transition Function: $\delta(q_i, a) = q_{(i+1) \bmod K}$, $\delta(q_i, b) = q_i$, $\delta(q_i, c) = q_i$



2b)



$$4) L_5 = \{ a^n b c^{2n} \mid n \geq 0 \}$$

Let us assume that for contradiction that L_5 is regular, then there exists a DFA $M = (Q, \Sigma, \delta, s_0, F)$ with K states such that it accepts L_5 .

Now, let us consider $K+1$ strings such that it contains $\epsilon, a, a^2, a^3, \dots, a^K$, which means that $K+1$ strings but has only K states, so by pigeonhole principle, we can say that there exist integers $i < j$ such that $\delta(s_0, a^i) = \delta(s_0, a^j)$.

Now let us assume an integer z , such that:

$$z = b c^{2i};$$

Now:

$$a^i z = a^i b c^{2i}$$

Now, we can see that the string

has twice as many c 's as compared to a , so that means it is a part of L_5 , hence, the DFA M must accept it.

Now, since M is in the same state after reading a^i and a^j , we can assume that it must behave identically when we are reading the same suffix z , hence it must also accept:

$$a^j z = a^j b c^{2i}$$

However, this string is not in L_5 , since it has j occurrences of a but only $2i$ occurrences of c , and since $j > i$, we now have $2i \neq 2j$. Thus, the number of c 's is not twice the number of a 's.

Hence, it is a contradiction and our assumption that L_5 is regular must be false as well.

Thus, L_5 is not regular.

5) Let us try to prove this by contradiction, so let us assume that L_6 is regular.

Now, let us consider that there exists a DFA such that $M = (Q, \Sigma, \delta, s_0, F)$ with $K = |Q|$ states that accepts L_6 .

Let us consider $K+1$ strings such that it contains strings like $a^{2^0}, a^{2^1}, a^{2^2}, \dots, a^{2^K}$, which means that there are $K+1$ strings but has only K states, now by Pigeonhole principle there exists integer i and j such that $i < j$, such that $\delta(s_0, a^{2^i}) = \delta(s_0, a^{2^j})$.

Now, let us assume $z = a^{2^i}$, then $a^{2^i} z = a^{2^i} \cdot a^{2^i} = a^{2^{i+1}}$.

Now, since 2^{i+1} is a power of 2, this string belongs to L_6 , so M must accept it.

Now, since M is in the same state after reading a^{2^i} and a^{2^j} , it means that it must also accept that $a^{2^j} z = a^{2^i} \cdot a^{2^j} = a^{2^j + 2^i}$

However, when $i < j$, the number $2^j + 2^i$ is not a power of 2. Indeed,

$$2^j + 2^i = 2^i (2^{j-i} + 1)$$

and the factor $2^{j-i} + 1$ is an odd integer which is greater than 1. Therefore, the product has an odd factor that is greater than 1, so we can say that it cannot be a power of 2, thus $a^{2^j + 2^i} \notin L_0$.

Hence, it is a contradiction, thus our assumption that L_0 is regular must be false.